

Machine Learning Project

Topic: Student Academic Performance Prediction Using Machine Learning

1: Introduction

- Background
 - Concepts
 - Statistical Numbers
- Problem Statement
- One related work

Background

Education plays a critical role in individual and societal development. However, predicting student academic performance remains a challenge due to the influence of multiple factors such as study habits, family background, social environment, and school-related conditions. With the advancement of machine learning, it is now possible to analyze these factors and build predictive models that can help identify students at risk of poor academic performance. This allows educators and institutions to take early intervention measures and improve overall student outcomes.

Concepts

- **Machine Learning:** A method that enables computers to learn patterns from data and make predictions without being explicitly programmed.
- **Supervised Learning:** A type of machine learning where the model is trained using labeled data (input and known output).
- **Classification:** A task where the model predicts a category (e.g., pass or fail).
- **Features:** Input variables used to make predictions (e.g., study time, age, absences).
- **Target Variable:** The output we want to predict (e.g., pass/fail).
- **Overfitting:** When a model learns the training data too well and performs poorly on new data.
- **Cross-validation:** A technique used to evaluate model performance more reliably.

Statistical Numbers

Student academic performance is influenced by several measurable factors. For example, attendance, study time, and parental education level have been shown to impact performance outcomes. In many educational datasets, patterns emerge showing that students with higher study time and lower absenteeism tend to achieve better grades. Identifying such patterns through data analysis allows for more informed academic support strategies.

Problem Statement

The purpose of this project is to leverage machine learning techniques to predict student academic performance and identify students at risk of failure. By analyzing factors such as study habits, attendance, and prior performance, the model provides early insights that can support timely intervention by educators. This helps improve student outcomes, reduce academic failure rates, and promote more effective and data-driven educational strategies.

Related Work

Previous studies have applied machine learning techniques to predict student performance using datasets containing demographic, academic, and behavioral features. Models such as Decision Trees, Logistic Regression, and Random Forest have been widely used. These studies found that factors such as study time, past academic performance, and attendance significantly influence student outcomes, demonstrating the effectiveness of machine learning in educational prediction tasks.

Cortez, P., & Silva, A. (2008). *Using Data Mining to Predict Secondary School Student Performance*. Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008).

Cortez & Silva (2008) applied data mining techniques to predict student performance using the Student Performance Dataset

Their study demonstrated that prior academic performance and study-related factors strongly influence final outcomes

This work serves as a foundation for applying machine learning to early academic risk detection

2: Dataset

- Basic Information
 - Reference
 - Number of rows and columns
 - Format
 - Meaning
- Detailed Information
 - Meaning of each column
 - Distribution
 - Correlation

Basic Information

- **Reference:** Kaggle
- **Dataset Name:** Student Performance Dataset
- **Source Link:** <https://www.kaggle.com/datasets/devansodariya/student-performance-data>
- **Number of Rows and Columns:** 395 rows and 33 columns
- **Format:** CSV (Comma-Separated Values)
- **Meaning:**

This dataset contains student-related information, including demographic details, academic performance, and behavioral factors. It is designed to analyze how different attributes, such as study time, attendance, and past performance, influence students' academic outcomes.

Detailed Information

Meaning of Each Column (Key Features)

S/N	Column	Meaning
1.	school	Type of school (GP OR MS)
2.	sex	Gender of Student
3.	age	Age of Student
4.	address	Address of Student (urban or rural)
5.	famsize	Size of Family
6.	Pstatus	Parents' cohabitation status (living together or apart)
7.	Medu	Mother's education level (ordered categorical levels from 1 - 5)
8.	Fedu	Father's education level (ordered categorical levels from 1 - 5)

9.	Mjob	Occupation of Mother
10.	Fjob	Occupation of Father
11.	reason	Reason for choosing the school
12.	guardian	Student's guardian
13.	traveltime	Home-to-school travel time (categorized into intervals)
14.	studytime	Weekly study time (ordered categorical levels from 1 - 5)
15.	failures	Number of past failures
16.	schoolsup	Extra educational support from school
17.	famsup	Family educational support
18.	paid	Extra paid classes within the course subject
19.	activities	Participation in extracurricular activities
20.	nursery	Attended nursery school
21.	higher	Intention to pursue higher education
22.	internet	Internet access at home
23.	romantic	Whether the student is in a romantic relationship
24.	famrel	Quality of family relationship (ratings from 1 - 5)
25.	freetime	Free time after school (ratings from 1 - 5)
26.	goout	How often a student goes out (ratings from 1 - 5)
27.	Dalc	Workday alcohol consumption (ratings from 1 - 5)
28.	Walc	Weekend alcohol consumption (ratings from 1 - 5)
29.	health	Current health status (ratings from 1 - 5)
30.	absences	Number of absences
31.	G1	First test score
32.	G2	Second test score
33.	G3	Final score (target basis)

Distribution

The dataset allows analysis of:

- Distribution of final grades (G3)
- Distribution of study time among students
- Frequency of absences
- Proportion of students who pass or fail

Expected observations:

- Most students fall within mid-grade ranges
- A smaller number of students perform extremely high or low
- The pass rate is typically higher than the fail rate

Correlation

Analysing Correlation will help identify relationships between variables.

Key expected relationships:

- Study time: positively correlated with performance
- Failures: negatively correlated with performance
- Absences: negatively correlated with performance
- Previous grades (G1, G2): strongly positively correlated with final grade (G3)

Model:

Logistic Regression: baseline model

K-Nearest Neighbors (KNN): comparison model

Decision Tree: interpretable model

Random Forest: main model

Achieve:

The project preprocesses the student dataset by handling categorical variables, scaling numerical features where necessary, and converting the final grade (G3) into a binary target variable (pass/fail).

After preprocessing, traditional machine learning models such as Logistic Regression, KNN, Decision Tree, and Random Forest are trained to classify whether a student will pass or fail.

Finally, the models are evaluated using accuracy, precision, recall, F1-score, and K-fold cross-validation to determine the best-performing model.